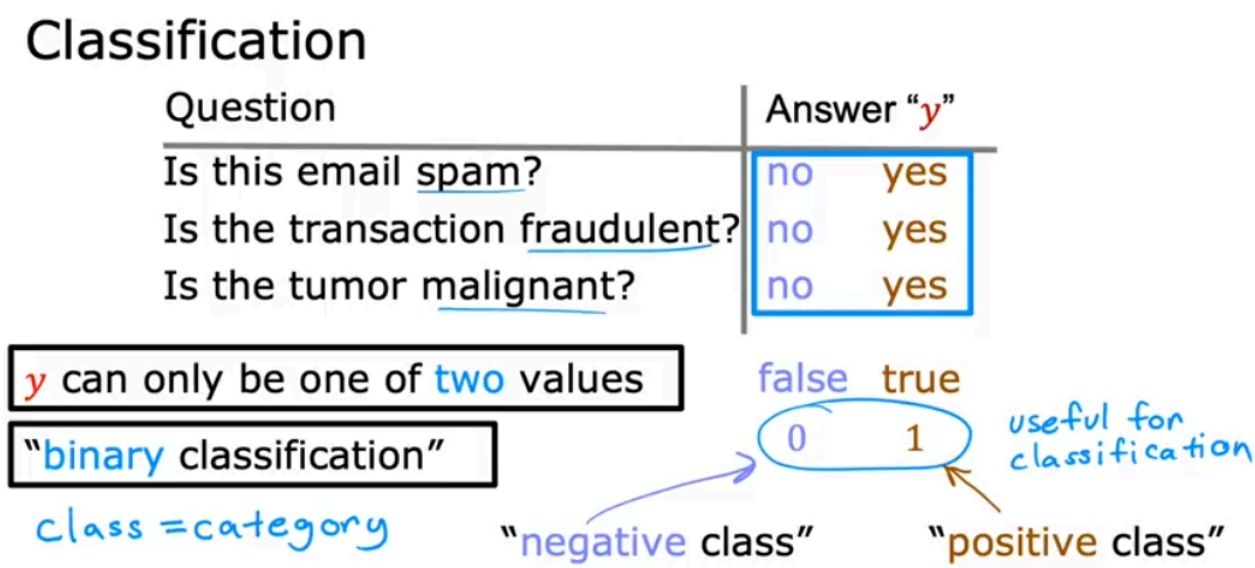
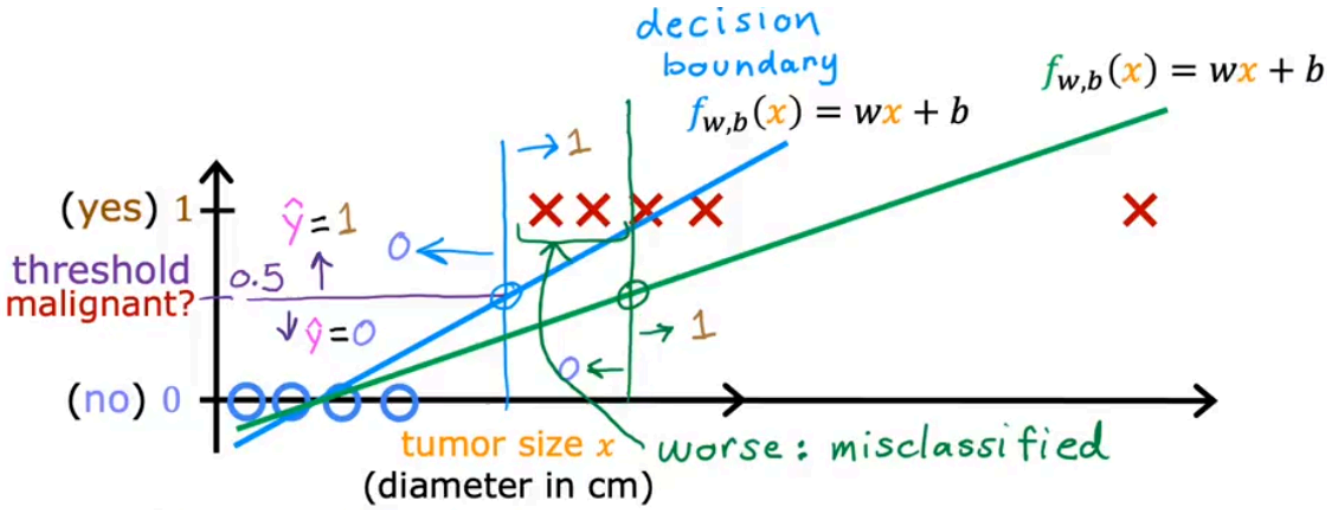
Day 8. Classification

Motivations



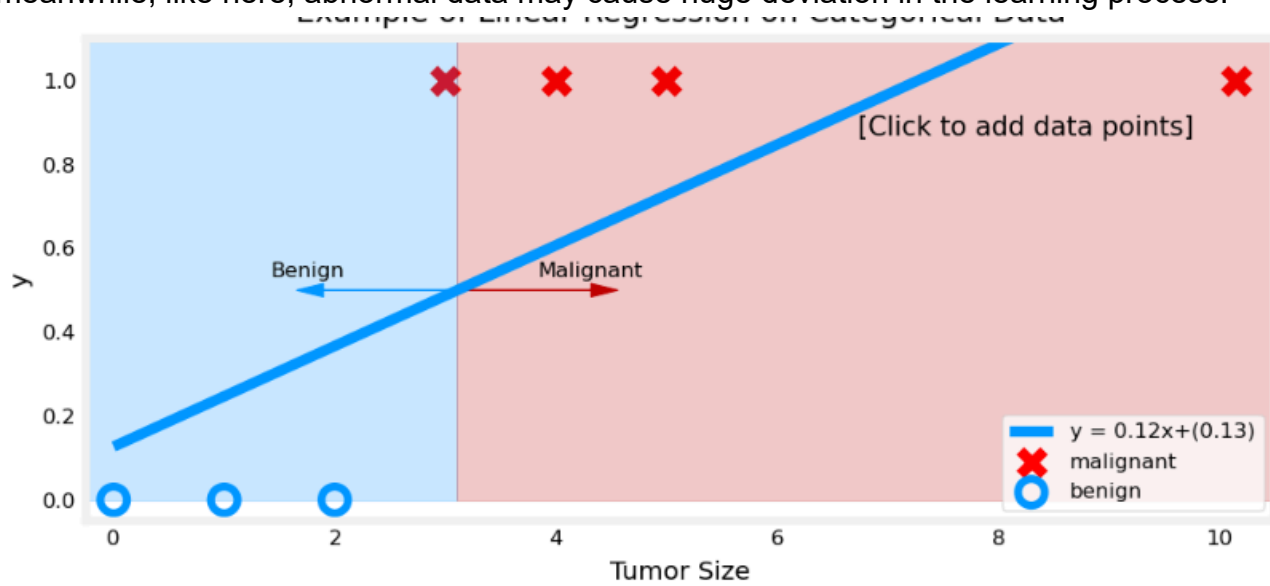
linear regression is not a good solution for classification, since in classification we are dealing with discrete results, while linear regression is a good solution to fit a continues function.



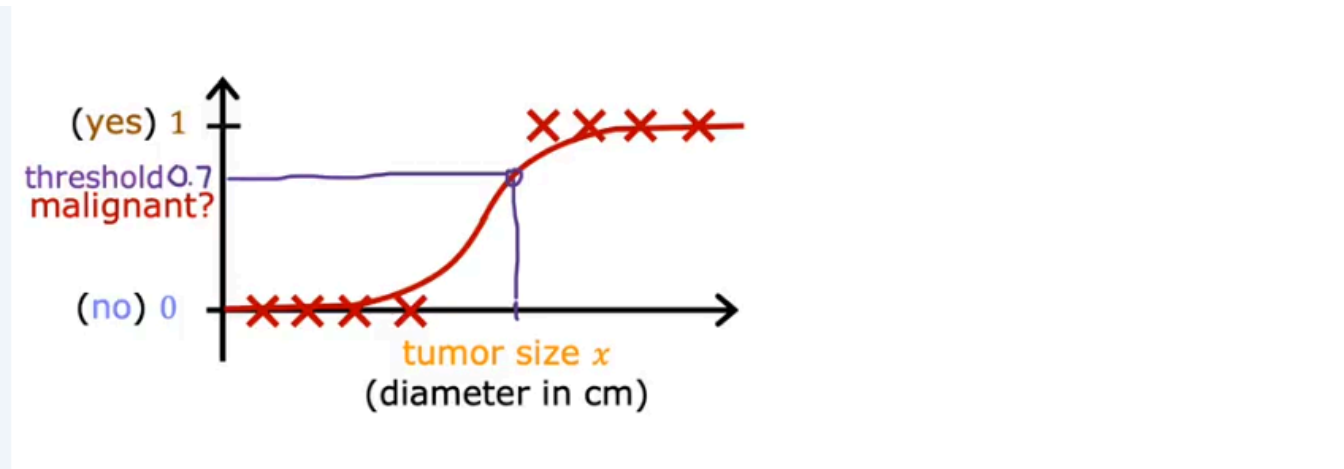
if $f_{w,b}(x) < 0.5 \rightarrow \hat{y} = 0$

if $f_{w,b}(x) \geq 0.5 \rightarrow \hat{y} = 1$

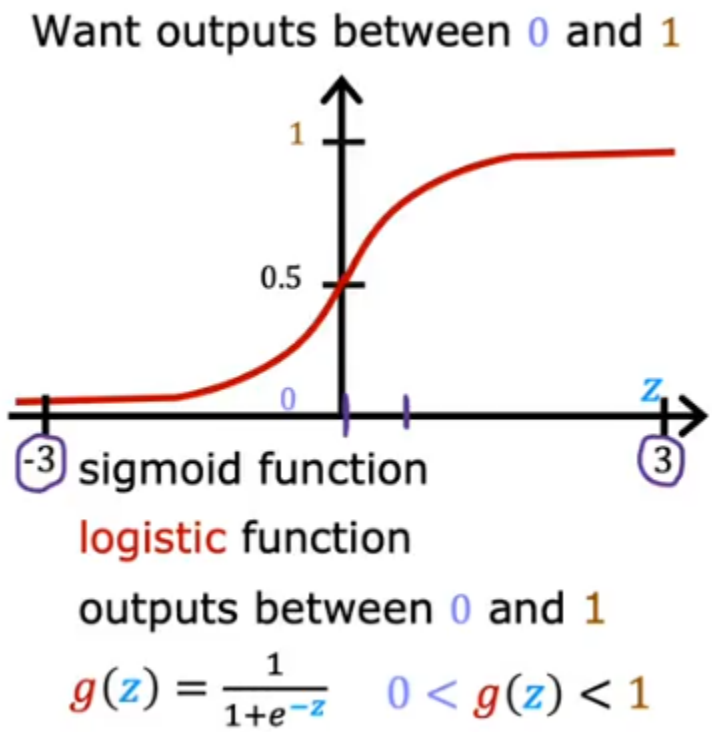
meanwhile, like here, abnormal data may cause huge deviation in the learning process.



Logistic regression

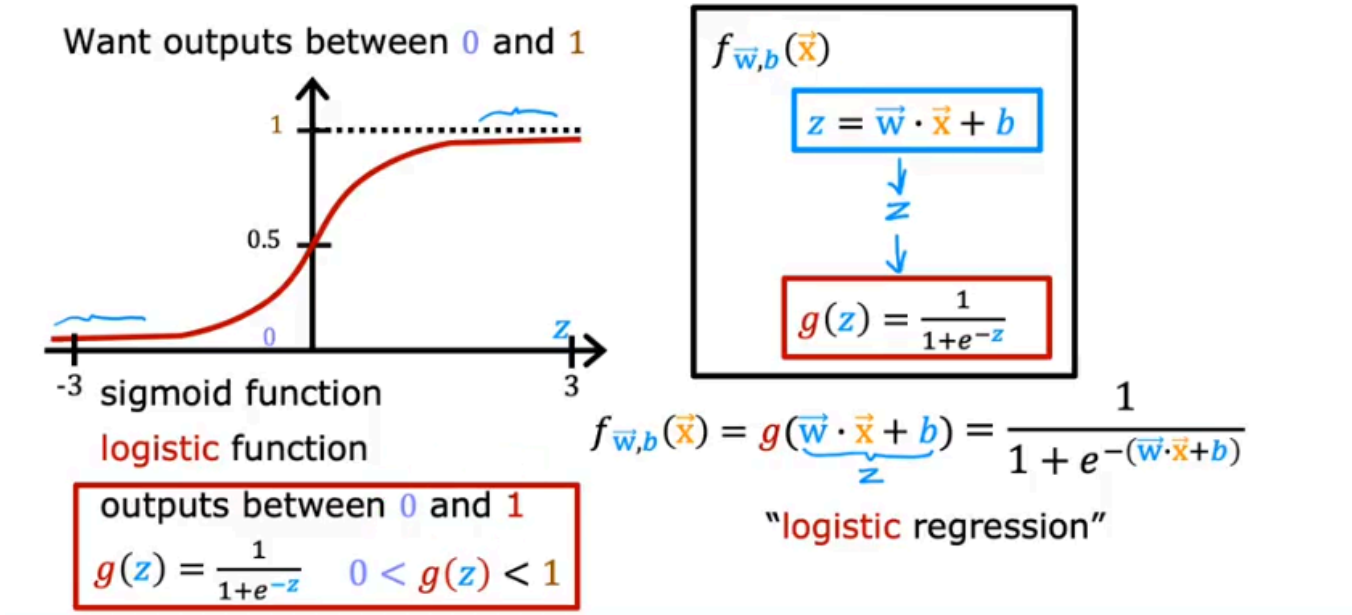


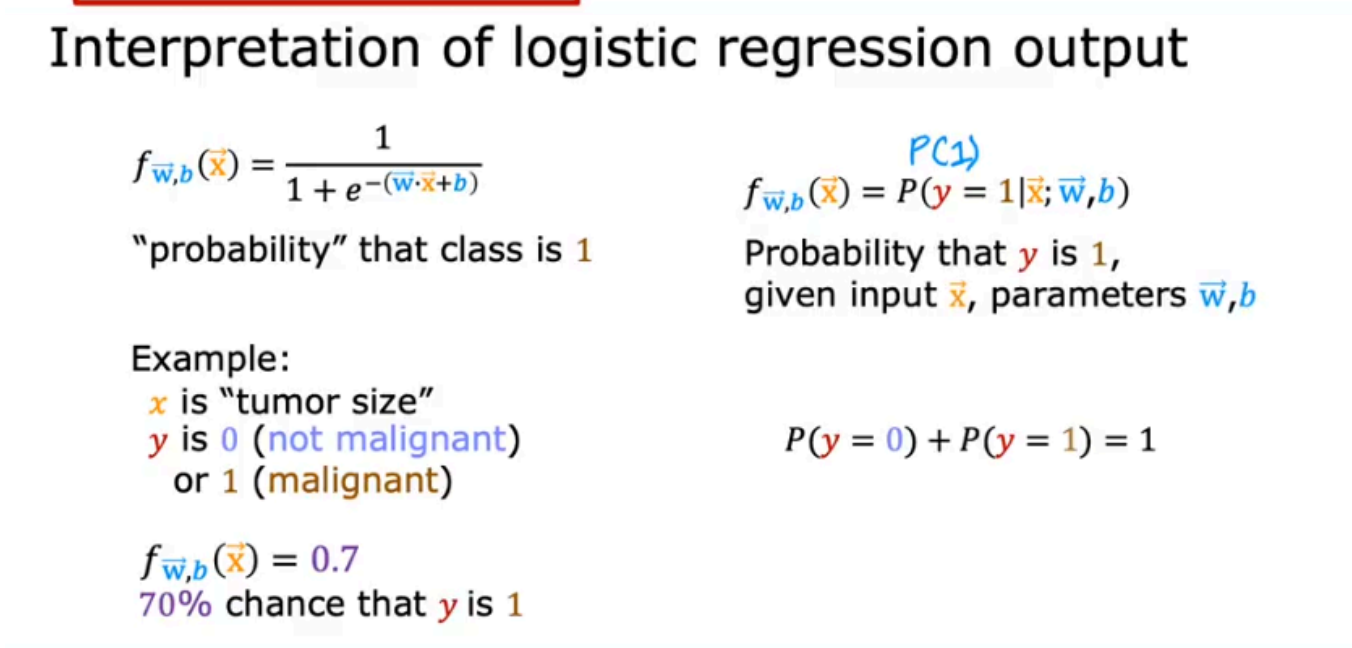
Sigmoid function (logistic function)



$$f(x) = \frac{1}{1 + e^{-x}}$$

> why sigmoid? and is there a better choice?

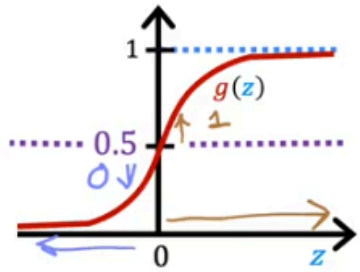




Decision boundary

to get a better sence how the logistic regression is calculating these predictions.

$$f_{\vec{x},b}(\vec{x}) = g(\vec{x} \cdot \vec{x} + b) = \frac{a}{e^{-(\vec{w}\cdot\vec{x}+b)}}$$



$$f_{\vec{w},b}(\vec{x}) = g(\underbrace{\vec{w}\cdot\vec{x}+b}_{z}) = \frac{1}{1+e^{-(\vec{w}\cdot\vec{x}+b)}}$$

$$= P(y=1|x;\vec{w},b) \quad 0.7 \quad 0.3$$

0 or 1? threshold

Is $f_{\vec{w},b}(\vec{x}) \geq 0.5$?

Yes: $\hat{y} = 1$ No: $\hat{y} = 0$

When is $f_{\vec{w},b}(\vec{x}) \geq 0.5$?

$$g(z) \geq 0.5$$

$$z \geq 0$$

$$\vec{w}\cdot\vec{x}+b \geq 0 \qquad \vec{w}\cdot\vec{x}+b < 0$$

$$\hat{y} = 1 \qquad \hat{y} = 0$$

$f_{\vec{w},b}(\vec{x})$

$$\boxed{z = \vec{w}\cdot\vec{x}+b}$$

$$\downarrow$$

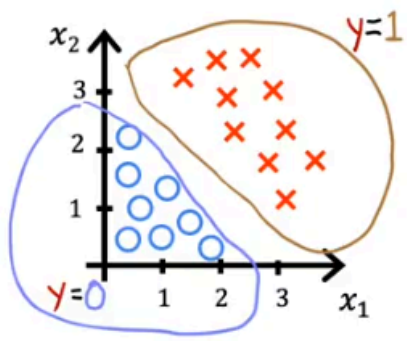
$$z$$

$$\downarrow$$

$$\boxed{g(z) = \frac{1}{1+e^{-z}}}$$

Decision boundary

$$f_{\vec{w},b}(\vec{x}) = g(z) = g(\underset{1}{w_1}x_1 + \underset{1}{w_2}x_2 + \underset{-3}{b})$$



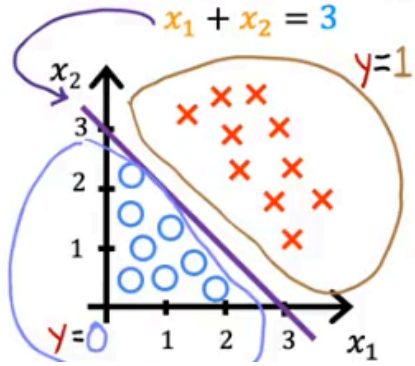
Decision boundary $z = \vec{w}\cdot\vec{x}+b = 0$

$f_{\vec{w},b}(\vec{x}) = g(z) = g(\underset{1}{w_1}x_1 + \underset{1}{w_2}x_2 + \underset{-3}{b})$

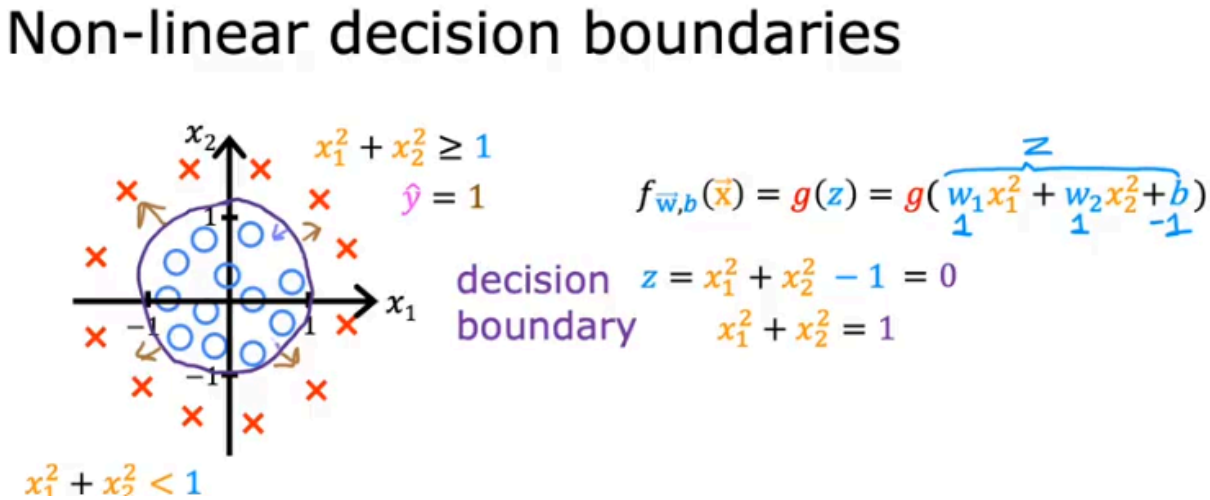
Decision boundary $z = \vec{w}\cdot\vec{x}+b = 0$

$$z = x_1 + x_2 - 3 = 0$$

$$x_1 + x_2 = 3$$



of course, we can have non-linear decision boundaries.



keywords

- logistic regression
- sigmoid function
- threshold
- decision boundary

After-class question

1. why sigmoid?
2. why sigmoid represents the probability?
3. how does the linear function $\vec{w} \times \vec{x} + b$ works?
4. how to decide the form of the polynomial term in regression?